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Evaluating the capability of the flux-limiter schemes in capturing strong shocks and discontinuities

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Abstract. A numerical study is conducted to investigate the capability of the flux-limiter TVD schemes in capturing sharp discontinuities like shock waves. For this purpose, four classical test problems are considered such as slowly moving shock, gas Riemann problem with high density and pressure ratios, shock wave interaction with a density disturbance and shock-acoustic interaction. The governing equations consist of one-dimensional and quasi-one-dimensional Euler equations solved using an in-house numerical code. In order to validate the solution, the obtained results are compared with other results found in the literature.

Keywords: TVD schemes, compressible flow, flux-limiter scheme, shock wave, nozzle

1. Introduction

The shock wave leads to a strong discontinuity in the fluid properties and high gradients of velocity and pressure. The flow across a shock wave is a non-isentropic process and the dissipation effects cannot be negligible. A great number of numerical schemes have been devised for the simulation of compressible gas dynamics during last several decades. The main difficulty in these simulations is to capture shocks and contact discontinuity. The pioneer works in this area were treated by Von Neumann and Richtmyer [1], Lax [2], Godunov [3], Lax and Wendroff [4], Van Leer [5, 6], Beam and Warming [7], Wilkins [8], Ben-Artzi and Falcovitz [9], Colella and Woodward [10], LeVeque [11] and Ogata and Yabe [12].

During the last decades, many shock-capturing schemes have been proposed that work with adequate robustness. Some of these schemes are the Total Variation Diminishing (TVD) schemes [13–15], approximate Riemann method [16], Essentially Non-Oscillatory (ENO) schemes [17,18], Weighted Essentially Non-Oscillatory (WENO) schemes [19], Flux-Vector Splitting (FVS) and Flux-Difference Splitting (FDS) schemes [20–22] and monotonicity preserving (MP) schemes [23,24].

Shock-capturing schemes have been investigated extensively by many researchers in recent years. For example, Tenaud et al. [25] investigated the capability of a large set of shock-capturing schemes to recover the basic interactions between acoustic, vorticity and entropy in a direct numerical simulation (DNS) framework. They considered four classes of shock-capturing schemes including TVD, MUSCL, Compact and ENO schemes. They compared the

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results obtained by using these methods with data obtained by fully resolved simulations (FRS) based on a sixth-order accurate Hermitian scheme. Daru and Tenaud [26] evaluated the accuracy of several high resolution TVD schemes in solving complex unsteady viscous shocked flows. They considered two types of discretization, a combined time and space (CTS) discretization and an independent time and space (ITS) discretization. They reviewed accuracy properties of each scheme on inviscid 1D and 2D test cases. They also studied the application of different schemes to the flow produced by the interaction of a reflected shock wave with the incident boundary layer in a shock tube. Their calculations are performed for two values of the Reynolds number, Re = 200 and 1000. Ren [27] proposed a robust finite volume shock-capturing scheme based on the rotated Roe's approximate Riemann solver. He presented several test cases to validate the proposed scheme. He found that the robustness of the rotated shock-capturing scheme is closely related to the way in which the direction of upwind differencing is determined. Petti and Bosa [28] presented an Eulerian accurate 2D shock-capturing method, based on weighted essentially non-oscillatory (WENO) and weighted average flux (WAF) schemes, to solve an advection dominated flow problem. The main aim of their model was to join these two techniques to reduce numerical dispersion, especially for long time simulations, by increasing accuracy in time and space of the advective solution. They examined their model for a number of test cases. The results of these tests verified the shock-capturing capabilities of the employed scheme, which permitted to solve extreme problems as water or pollutant shocks. Tu et al. [29] developed a new shock-capturing method based on upwind schemes and flux-vector splittings. They examined some numerical cases to evaluate their shock-capturing method. They reported that their method can sufficiently suppress non-physical oscillations near shock waves and also prevent some non-physical solutions such as over expansions. In addition, the limiters can largely maintain the resolution power of the original linear schemes and have little influence on the accuracy of them. Baghlani et al. [30] developed a robust and accurate high-resolution finite-volume scheme which employs flux-vector splitting (FVS) scheme. They accomplished an estimation of flux gradients at the cell interface by means of a flux-limiter in the characteristic field to increase the accuracy of the FVS and suppress excessive numerical dissipation with it. They tested this method for hydraulic jump, 1D dam break and 2D dam break problems, and reported that their results show very satisfactory agreement with experiments, analytical solution and other numerical schemes. Bogey et al. [31] developed a shock-capturing methodology based on an adaptive spatial filtering for high-accuracy non-linear computations including low-dissipation time integration and centered space differencing. They derived a shock-detection procedure based on a Jameson-like shock sensor to apply the shock-capturing filtering only around shocks. They reported that this methodology is capable of capturing shocks without providing dissipation outside shocks. Galiano and Zapata [32] developed a new flux-limiter method based on the Richtmyer two-step Lax-Wendroff (R2LW) method coupled with a conservative upwind method and a nonconventional flux-limiter function. Their proposed method is TVD stable and preserves the linear stability condition of the R2LW method. They reported that their new method provides accurate results for nonlinear hyperbolic equations with discontinuous solutions. Li et al. [33] evaluated the performance of generalized Riemann problem (GRP) scheme and gas-kinetic scheme (GKS), which are two high resolution shock capturing schemes for fluid simulations, in many 1D and 2D flow computations including the large density ratio problem, highly expansion wave, and the blunt body problem. They reported that the GRP exhibits a slightly better computational efficiency, and has comparable accuracy with GKS for the Euler solutions in 1D case, but the GKS is more robust than GRP. For the 2D high Mach number flow simulations, the GKS is absent from the shock instability and converges to the steady state solutions faster than the GRP.

In the present work, the shock capturing capability of the flux-limiter TVD schemes is evaluated by applying these schemes to several test problems including slowly moving shock, gas Riemann problem with high density and pressure ratios, shock wave interaction with a density disturbance and shock-acoustic interaction. For the slowly moving shock case, Superbee, CHARM, H-QUICK, Minmod and van Leer limiters are examined. In three other cases, the solutions are obtained by using Superbee limiter. The results published in the literature are also presented to validate the solution.

2. Governing equations and numerical schemes

2.1. Governing equations

The test problems considered in this study are solved from the One-dimensional and the quasi-one-dimensional Euler equations.

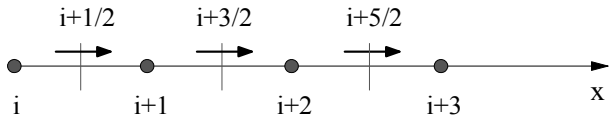


Fig. 1. Schematic view of grid and related nomenclature.

2.1.1. One-dimensional Euler equations

The One-dimensional Euler equations can be expressed as [34]

$$\frac{\partial Q}{\partial t}+\frac{\partial F}{\partial x}=0 \tag{1}$$

where Q and F are the column vectors representing the variable vector (vector of conservative variables) and the flux vector, respectively, which are defined as

$$Q=\begin{bmatrix}\rho\\ \rho u\\ E\end{bmatrix},\quad F=\begin{bmatrix}\rho u\\ \rho u^2+p\\ u(E+p)\end{bmatrix} \tag{2}$$

where ρ, u and p are density, velocity and pressure, respectively. E is the total energy and for a perfect gas ($\gamma = 1.4$) is given by

$$E=\frac{p}{\gamma-1}+\frac{1}{2}\rho u^2 \tag{3}$$

2.1.2. Quasi-one-dimensional Euler equations

The Quasi-one-dimensional Euler equations can be written as [34]

$$\frac{\partial Q}{\partial t}+\frac{\partial F}{\partial x}=G \tag{4}$$

where Q and F are the variable vector and flux vector, respectively, which are obtained from Eq. (2), and G is the source vector, which is computed as

$$G=\begin{bmatrix}-\rho u\frac{1}{S}\frac{\partial S}{\partial x}\\ -\rho u^2\frac{1}{S}\frac{\partial S}{\partial x}\\ -\frac{u(E+p)}{S}\frac{\partial S}{\partial x}\end{bmatrix} \tag{5}$$

where $S = S(x)$ is the cross-sectional area.

2.2. Numerical schemes

2.2.1. Roe scheme

In the Roe scheme, the solution is based on solving a localized Riemann problem to calculate the flux at a given face of the domain. The governing equations (Eq. (4)) are discretized using a forward difference scheme for time derivative term and a central difference scheme for space derivative term as follows

$$\frac{Q_i^{n+1}-Q_i^n}{\Delta t}+\frac{F_{i+1/2}^n-F_{i-1/2}^n}{\Delta x}=G_i^n \tag{6}$$

The above equation can be rewritten as

$$Q_i^{n+1}=Q_i^n-\frac{\Delta t}{\Delta x}\left(F_{i+1/2}^n-F_{i-1/2}^n\right)+G_i^n\Delta t \tag{7}$$

The nomenclature of grid points and faces are presented schematically in Fig. 1. Matrices Q and F are computed for each cell while $F_{i+1/2}^n$ and $F_{i-1/2}^n$ are fluxes through cell faces and computed as [35]

$$F^n_{i+1/2} = \frac{1}{2}\left(F^n_{i+1} + F^n_i\right) - \frac{1}{2}\left|\bar{A}_{i+1/2}\right|\left(Q^n_{i+1} + Q^n_i\right) \tag{8}$$

where A is the flux Jacobian matrix which can be diagonalized as

$$\left|\bar{A}\right| = \bar{A}^+ - \bar{A}^- = \bar{R}\left|\bar{\Lambda}\right|\bar{L} \tag{9}$$

where R, Λ and L are the right eigenvector matrix, the eigenvalue matrix and the left eigenvector matrix, respectively. The eigenvalue matrix, Λ, is a diagonal matrix of the wavespeeds

$$\Lambda = \begin{bmatrix} u-c & 0 & 0 \\ 0 & u & 0 \\ 0 & 0 & u+c \end{bmatrix} \tag{10}$$

The corresponding eigenvector matrix, R, and its inverse, L, are given by

$$R = \begin{bmatrix} 1 & 1 & 1 \\ u-c & u & u+c \\ H-uc & u^2/2 & H+uc \end{bmatrix}, \quad L = \begin{bmatrix} \frac{1}{2}(b_1+u/c) & -\frac{1}{2}(ub_2+1/c) & b_2/2 \\ 1-b_1 & b_2 u & -b_2 \\ \frac{1}{2}(b_1-u/c) & -\frac{1}{2}(ub_2-1/c) & b_2/2 \end{bmatrix} \tag{11}$$

where H denotes the total enthalpy and the variables b_1 and b_2 are defined as

$$H = \frac{P\gamma}{(\gamma-1)\rho} + \frac{u^2}{2}, \quad b_1 = \frac{\gamma-1}{2}\left(\frac{u}{c}\right)^2, \quad b_2 = \frac{\gamma-1}{c^2} \tag{12}$$

In Eqs (8) and (9), the overbar denotes Roe-averaged quantities which are obtained from the following equations [35]

$$\bar{\rho} = \sqrt{\rho_L \rho_R}, \quad \bar{u} = \frac{\sqrt{\rho_L}u_L + \sqrt{\rho_R}u_R}{\sqrt{\rho_L}+\sqrt{\rho_R}}, \quad \bar{H} = \frac{\sqrt{\rho_L}H_L + \sqrt{\rho_R}H_R}{\sqrt{\rho_L}+\sqrt{\rho_R}} \tag{13}$$

where subscripts R and L represent the right and left cells of each face respectively.

2.2.2. Flux-limiter scheme

The main idea of developing flux-limiter schemes is to prevent the spurious numerical oscillations and to ensure that the results are physically realistic. In this study, a flux-limiter scheme equipped with Superbee [36,37], CHARM [38], H-QUICK [39], Minmod [37] and van Leer [5] limiters is used to limit the solution gradient near shocks or discontinuities.

The corresponding flux in this scheme is a combination of a low-order flux formula and a higher-order formula as follows

$$F = F^L + \Phi(\theta)(F^H - F^L) \tag{14}$$

Here, the high-order flux F^H is given by the Lax-Wendroff method whereas the low-order flux F^L is computed by using the Roe method. $\Phi(\theta)$ is the flux-limiter function and θ is the ratio of successive gradients indicating the smoothness of the solution. The flux-limiter functions employed in the present study are given in Eq. (15).

$$\Phi(\theta) = \max\left[0,\ \min(1,\ 2\theta),\ \min(2,\ \theta)\right] \text{ Superbee} \tag{15-1}$$

$$\Phi(\theta) = \begin{cases} \theta(3\theta+1)/(\theta+1)^2 & \theta > 0 \\ 0 & \theta \leqslant 0 \end{cases} \text{ CHARM} \tag{15-2}$$

$$\Phi(\theta) = 2(\theta+|\theta|)/(\theta+3) \text{ H-QUICK} \tag{15-3}$$

$$\Phi(\theta) = \max\left[0,\ \min(1,\ \theta)\right] \text{ Minmod} \tag{15-4}$$

$$\Phi(\theta) = (\theta+|\theta|)/(1+|\theta|) \text{ van Leer} \tag{15-5}$$

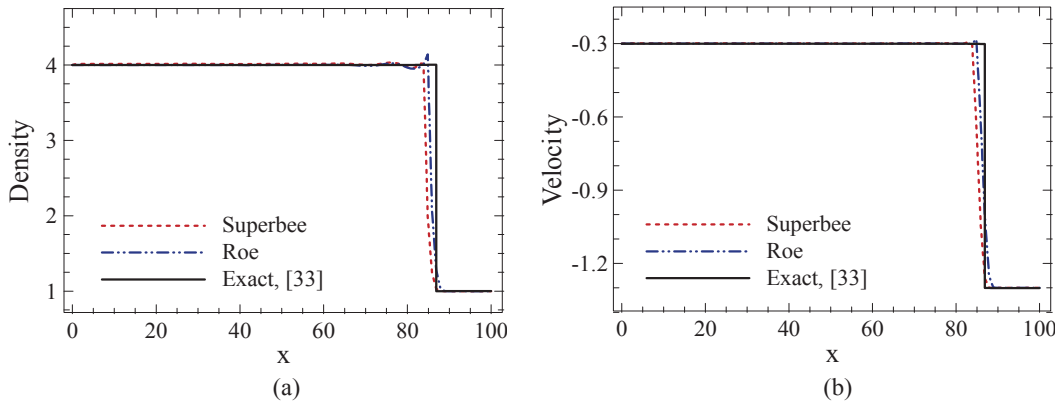


Fig. 2. (a) Density distribution (b) velocity distribution; $t = 2000$.

The ratio $\theta_{i+1/2}$ can be considered as a measure of the smoothness of the data near $x_{i+1/2}$, which is defined as

$$\theta_{i+1/2} = \begin{cases} (Q_i - Q_{i-1})/(Q_{i+1} - Q_i) & \text{for } \bar{u} > 0 \\ (Q_{i+2} - Q_{i+1})/(Q_{i+1} - Q_i) & \text{for } \bar{u} < 0 \end{cases} \tag{16}$$

It should be noted that the value of the limiter function, $\Phi(\theta)$, depends on the solution smoothness. $\Phi(\theta)$ is equal to zero for sharp gradient such that the flux is represented by a low-order formula while for smooth solution it is equal to 1 and so the flux is represented by a high-order formula.

3. Numerical results

In this section, four test problems including slowly moving shock, gas Riemann problem with high density and pressure ratios, shock wave interaction with a density disturbance and shock-acoustic interaction are considered for evaluation. These problems are widely used in literature to test the accuracy of shock-capturing schemes. The numerical results are obtained by an in-house numerical code equipped with the flux-limiter schemes. The variables for density, velocity and pressure in the numerical analysis are nondimensionalized.

3.1. Slowly moving shock

A slowly rightward-moving shock with the initial conditions of Eq. (17), which was studied in [33], is considered for this test case. The problem is solved using the current numerical code, which is equipped with many limiter schemes including Roe, Superbee, CHARM, H-QUICK, Minmod and van Leer. The computational domain is [0, 100], which contains 100 grid points, at time $t = 2000$.

$$\begin{cases} \rho = 4.0, \;\; u = -0.3, \;\; p = 4/3, \;\; \gamma = 5/3 \quad \text{for } \; 0 \leqslant x < 20 \\ \rho = 1.0, \;\; u = -1.3, \;\; p = 10^{-6}, \;\; \gamma = 5/3 \quad \text{for } \; 20 < x \leqslant 100 \end{cases} \tag{17}$$

The distributions of density and velocity at time $t = 2000$ are presented in Figs 2 and 3. The results obtained by the different limiter schemes and the exact solution presented by Li et al. [33] are plotted in these figures. The current problem clearly demonstrates the capability of these schemes in the capturing of shock waves. The shock is initially at $x = 20$ and at $t = 2000$, it is located at about $x = 87.8$. As it can be seen in these figures, there are some small non-physical oscillations before the shock front. In addition, the presented schemes cannot predict the shock location accurately. The Superbee limiter gives the least non-physical oscillations while the CHARM limiter shows the largest oscillations. Furthermore, the Roe scheme can get the most accurate position of the shock.

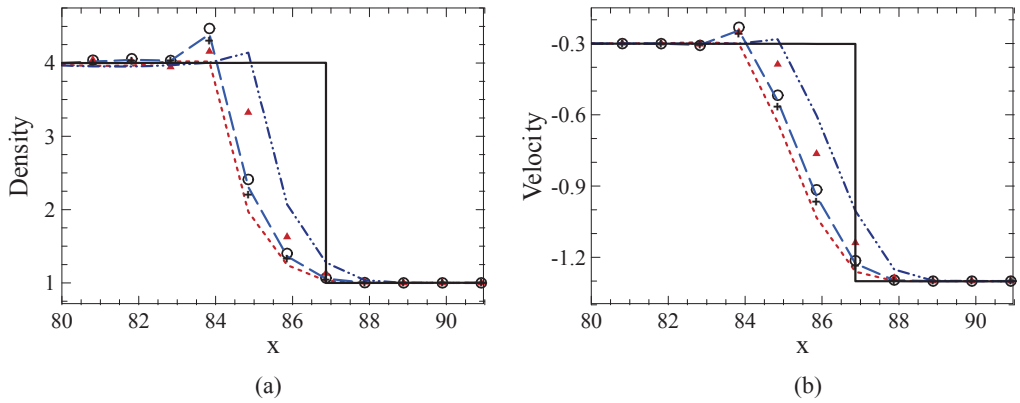


Fig. 3. Close-ups in the vicinity of the shock, $t = 2000$; (a) density distribution (b) velocity distribution; - - - - -: Superbee; ○: CHARM; − − −: H-QUICK; ▲: Minmod; +: van Leer; −·· − ··−: Roe; ——: exact solution [33].

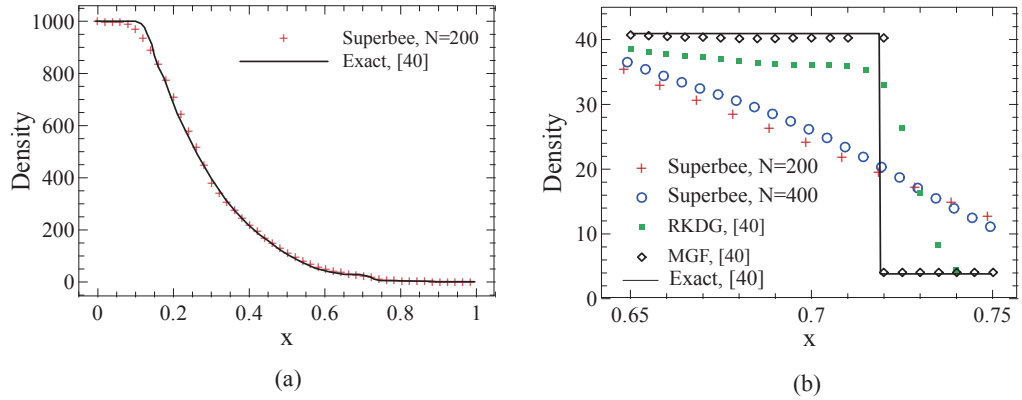


Fig. 4. Comparison of the Superbee flux-limiter scheme with literature data for 200 and 400 cells, $t = 0.15$; (a) density distribution (b) a closer view of the contact discontinuity.

3.2. *Gas Riemann problem with high density and pressure ratios*

In this test case, the governing equations are the one-dimensional Euler equations, which are solved using the current numerical code by employing the Superbee flux-limiter scheme. The computational domain is [0, 1]. This test case is treated in [40]. The results are presented and discussed at time $t = 0.15$ with 200 and 400 grid points. The initial conditions are

$$\begin{cases} \rho = 1000.0, \;\; u = 0.0, \;\; p = 1000.0, \;\; \gamma = 1.4 \quad \text{for } \; 0 \leqslant x < 0.3 \\ \rho = 1.0, \qquad u = 0.0, \;\; p = 1.0, \qquad \gamma = 1.4 \quad \text{for } \; 0.3 < x \leqslant 1 \end{cases} \tag{18}$$

Figures 4 and 5 show the results obtained by the Superbee flux-limiter scheme. The exact solution and the results of RKDG and MGF schemes reported by Chen and Zhou [40] are also presented here. However, the Superbee scheme gives the same trend of density variation but it is not robust enough to detect the small discontinuity. The solutions are non-oscillatory near the contact discontinuity. In addition, it is observed that the solution with 400 grid points shows slightly better performance near the discontinuity in the density.

3.3. *Shock wave interaction with a density disturbance*

This test case was proposed in [23]. The one-dimensional Euler equations on a computational domain [0, 10], consisting of 200 and 400 grid points, are solved using the current numerical code equipped with the Superbee

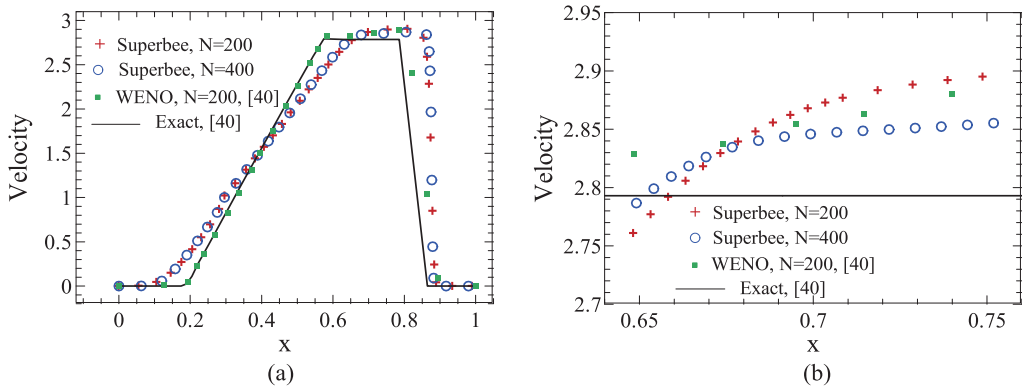


Fig. 5. (a) Velocity distribution (b) a closer view of the contact discontinuity; 200 and 400 cells, $t = 0.15$.

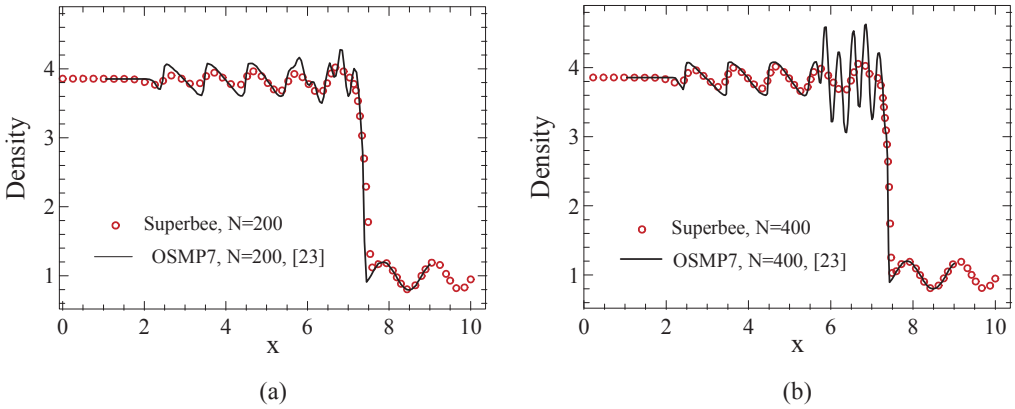


Fig. 6. Density distribution at $t = 1.8$; (a) 200 cells (b) 400 cells.

flux-limiter scheme. The shock is initially located at $x = 1.0$. The computational time is $t = 1.8$ and the problem is initialized with

$$\begin{cases} \rho = 3.857, & u = 2.629, \quad p = 10.3333, \quad \gamma = 1.4 \quad \text{for } 0 \leqslant x < 1 \\ \rho = 1.0 + 0.2\sin(5x), & u = 0.0, \quad p = 1.0, \quad \gamma = 1.4 \quad \text{for } 1 \leqslant x \leqslant 10 \end{cases} \tag{19}$$

The obtained results with 200 and 400 grid points are plotted in Fig. 6. The results from the OSMP7 scheme obtained by Daru and Tenaud [23] are also presented for comparison. They reported that the OSMP7 results with 400 grid points are very good for this test case. The Superbee flux-limiter scheme can predict the shock location on the fine and coarse grids very well, but it cannot detect the sinusoidal density distribution precisely. The computation on the fine grid cannot improve the numerical results sufficiently.

3.4. Shock-acoustic interaction

In this test problem, the steady inviscid compressible flow through a convergent-divergent nozzle, so called a de Laval nozzle (Fig. 7), is simulated by solving the quasi-one-dimensional Euler equations. The shock wave location and the distribution of fluid properties along the axis of the nozzle are determined. The cross-sectional area of the nozzle is a function of x and follows from Eq. (20). The computational domain is $[-10, 10]$, and the flow is transonic while a normal shock wave occurs at the downstream of the nozzle throat.

$$\begin{cases} S(x) = 1.0 - 0.661514\exp\left(-Ln(2)(x/0.6)^2\right) & \text{for } -10 \leqslant x \leqslant 0 \\ S(x) = 0.536572 - 0.198086\exp\left(-Ln(2)(x/0.6)^2\right) & \text{for } 0 < x \leqslant 10 \end{cases} \tag{20}$$

The boundary conditions are employed as follows:

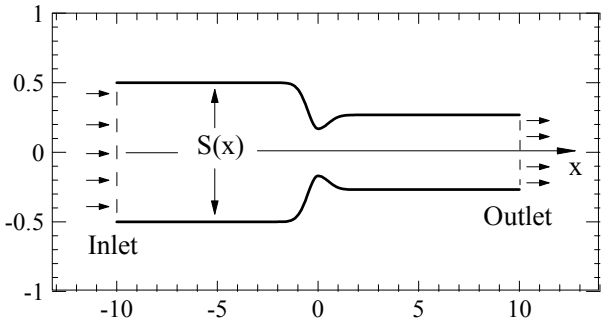


Fig. 7. Convergent-divergent nozzle.

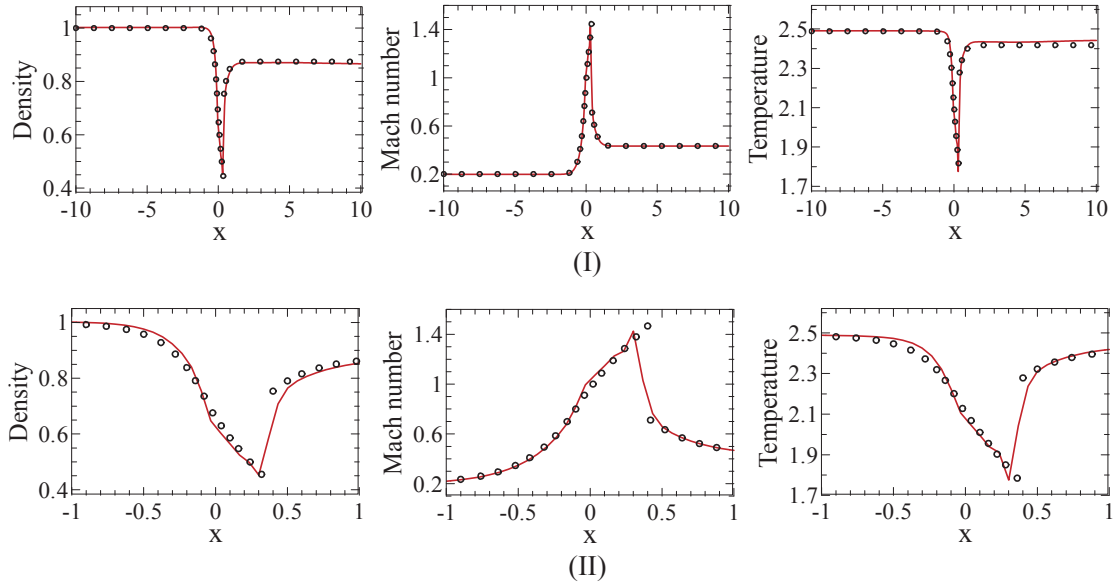


Fig. 8. Distributions of density, Mach number and temperature along the nozzle axis; ——: Superbee scheme, ○: analytical solution; (I) complete view, (II) detailed view in the vicinity of the shock wave.

- At the inlet: $M_{in} = 0.2006533$, $\rho_{in} = 1.0$ and $p_{in} = 1/\gamma$ where M is the Mach number and γ is the ratio of specific heats (γ is 1.4 for air).
- At the outlet: $p_{out} = 0.6071752$.

In addition, an analytical solution of the problem is performed to testify the numerical results. The location of the shock wave is initially unknown. For the analytical solution, the location of the shock is assumed at a point at the downstream of the throat. The distribution of the fluid properties along the x-axis is obtained by using the gas dynamics equations for the isentropic flow and the normal shock wave [41]. The computed output pressure is compared to its known value ($p_{out} = 0.6071752$) and if there is good agreement between them, the solution procedure will be stopped otherwise a new shock location is considered and the solution is repeated. On the other hand, a continuous trial and error process is done to obtain the best solution. The obtained analytical results are shown in Fig. 8. The analytically predicted location of shock wave is approximately at $x = 0.385$.

The distributions of density, Mach number and temperature are shown in Fig. 8. As it is observed from this figure, the results of the Superbee scheme are in good agreement with analytical results. The solutions are non-oscillatory near the shock wave, however, there are slight overshoots and undershoots near the shock location. Furthermore, the numerical scheme can get the relatively correct position of the shock ($x = 0.37$). This reveals that the Superbee flux-limiter scheme is quite capable to capture sharp discontinuities like shock waves.

4. Conclusions

The present study evaluates the capability of the flux-limiter TVD schemes in capturing shock waves and discontinuities. Four classical test cases are considered for this evaluation, which are slowly moving shock, gas Riemann problem with high density and pressure ratios, shock wave interaction with a density disturbance and shock-acoustic interaction. For the case of slowly moving shock, several limiters including Superbee, CHARM, H-QUICK, Minmod and van Leer are examined. In this test case, the numerical results show some small non-physical oscillations. The minimum of these oscillations can be observed for the Superbee limiter while the CHARM limiter shows the maximum oscillation amplitude. Furthermore, the Roe scheme can predict the most accurate position of the shock. In other cases, the solutions are obtained by using Superbee limiter. The presented test cases show that, however the Superbee scheme is not robust enough to detect the small discontinuity and there are very weak overshoot and undershoot near the shock wave or sharp discontinuities, but the obtained results are promising. It should be noted that no particular limiter has been found to work well for all problems, and a particular choice is usually made on a trial and error basis.

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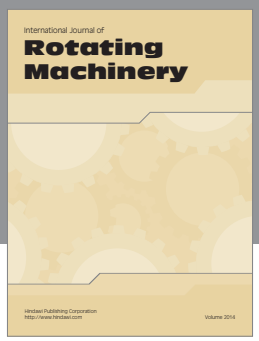
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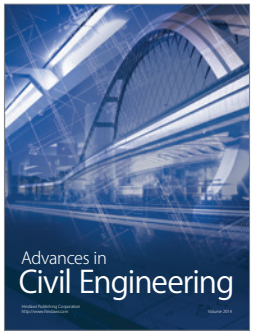


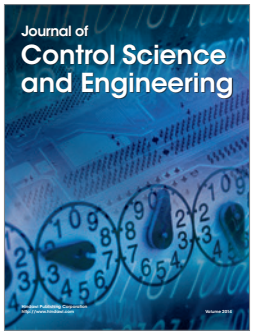






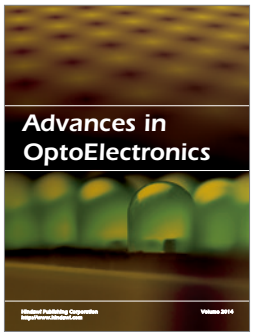






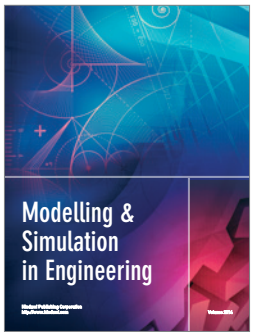






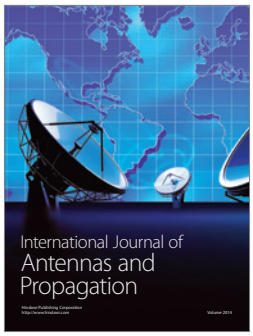




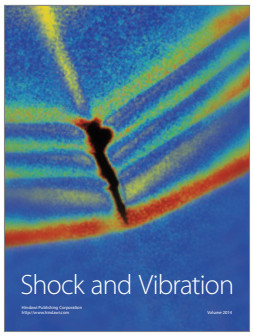


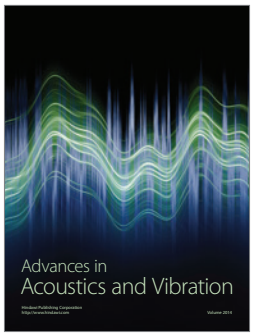












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